

URGENT: We ran out of 5-digit catalog numbers on 2026-07-11. The official USSF SATCAT is now at 100831. All newly cataloged objects have 6-digit catalog numbers of 100000+ and GP data will not be available for them using the TLE format. Please be sure to use one of Celestrak's other formats which remove the 5-digit catalog number limitation (and finally fix the Y2K problem) and to adhere to our usage policy when retrieving any Celestrak data. Follow @TSKelso on Bluesky for the latest updates, tutorials, and changes as we move forward. Spread the word! - TS

A New Way to Obtain GP Data (aka TLEs)

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Background

The US government has provided GP or *general perturbations* orbital data to the rest of the world since the 1970s. These data are produced by fitting observations from the US Space Surveillance Network (SSN) to produce Brouwer mean elements using the SGP4 or *Simplified General Perturbations 4* orbit propagator.

Many of you are familiar with this data in the form of TLEs or *Two-Line Element Sets*. TLEs were designed to provide the minimum data necessary to propagate the orbit of a resident space object (RSO) at a time when both bandwidth for transmission or digital storage were extremely limited. In fact, at the time, transmission might be via fax, hard copy (postal delivery), or even read over the phone and storage was handled using punch cards or magnetic tape.

While this format has served us well for many decades, it has not been without its share of problems. For example, the choice of a two-digit year caused many problems approaching Y2K—problems that were side-stepped by redefining what those two digits represented—but that Y2K problem persists fully 20 years into the 21st century. And now we are approaching another milestone where we will no longer be able to catalog all the objects we track within the 5-digit catalog number limitation of the TLE format.

One of the key drivers forcing us to consider tracking more than 100,000 objects is the activation of the Space Fence on Kwajalein Atoll. The Space Fence reached [initial operational capability \(IOC\) on 2020 Mar 27](#) and is expected to track far more than the ~26,000 objects currently tracked by the SSN—perhaps by as much as an order of magnitude.

And we are expecting to see public availability of data from the Space Fence starting some time this summer (2020). The 18th Space Control Squadron (18 SPCS) has already transitioned internally to using 9-digit catalog numbers in support of these changes and we expect 18 SPCS to release data from the Space Fence using 9-digit catalog numbers.

The Solution

Celestrak—working closely with [Space Track](#)—has begun making the GP data available via standard queries using the Orbit Mean-Elements Message (OMM) that is part of the Orbit Data Messages (ODM) Recommended Standard [CCSDS 502.0-B-3](#) developed by [The Consultative Committee for Space Data Systems \(CCSDS\)](#) in November 2009. We are recommending the XML format of Version 2.0 of the OMM, as defined in [XML Specification for Navigation Data Messages \(CCSDS 505.0-B-3\)](#) to ensure future compatibility and interoperability.

There are XML and KVN (key-value notation) versions of the OMM standard and Celestrak will provide all mandatory elements of those formats. Some elements may be blank (KVN) or null (XML), if not available via the current TLE format. An example might be that an object in the current analyst sat range (80000-series) typically will not have a name (OBJECT_NAME) or International Designator (OBJECT_ID).

Use of the new data queries is NOT required for most users at this time, since Celestrak will continue to provide data for RSOs with 5-digit catalog numbers in the TLE/3LE or 2LE formats. The current focus is to provide software developers a way to test modifications to their code to support using the new OMM format and 9-digit catalog numbers. Legacy links to fixed .txt files will continue indefinitely, although links on web pages will eventually be transitioned to use the new GP query and allow users to define their default format. Of course, TLE formats will not support objects with catalog numbers above 99999.

Additionally, data will be provided in both JSON and CSV formats, using the same keywords and definitions as provided in the OMM standard ([CCSDS 502.0-B-3](#), Table 4-1), although null/blank or redundant (e.g., CENTER_NAME = EARTH, REF_FRAME = TEME, TIME_SYSTEM = UTC, MEAN_ELEMENT_THEORY = SGP4) mandatory fields will not be included.

Celestrak will work to ensure that all GP data received via 18 SPCS and Space Track will be ingested in a way that supports users requesting GP data in any of the TLE or OMM formats.

The Implementation

All GP queries on Celestrak will take the form:

- <https://celestrak.org/NORAD/elements/gp.php?{QUERY}=VALUE&FORMAT=VALUE>

where {QUERY} is:

- CATNR: Catalog Number (1 to 9 digits). Allows return of data for a single catalog number.
- INTDES: International Designator (yyyy-nnn). Allows return of data for all objects associated with a particular launch.
- GROUP: Groups of satellites provided on the Celestrak Current Data page.
- NAME: Satellite Name. Allows searching for satellites by parts of their name.
- SPECIAL: Special data sets for:
 - The GEO Protected Zone (SPECIAL=GPZ)
 - GPZ Plus (SPECIAL=GPZ-PLUS)
 - Potential Decays (SPECIAL=DECAYING)

{QUERY} **must** be uppercase.

Allowed formats are:

- TLE or 3LE: Three-line element sets including 24-character satellite name on Line 0.
- 2LE: Two-line element sets (no satellite name on Line 0).
- XML: CCSDS OMM XML format including all mandatory elements.
- KVN: CCSDS OMM KVN format including all mandatory elements.
- JSON: OMM keywords for all GP elements in JSON format.
- JSON-PRETTY: OMM keywords for all GP elements in JSON pretty-print format.
- CSV: OMM keywords for all GP elements in CSV format.

The FORMAT specification is optional, but defaults to CSV (as of 2026 May 09).

Examples:

- TLE format for ISS (25544)
<https://celestrak.org/NORAD/elements/gp.php?CATNR=25544&FORMAT=TLE>
- KVN format for ISS (25544)
<https://celestrak.org/NORAD/elements/gp.php?CATNR=25544&FORMAT=KVN>
- XML format for the Stations list found on Celestrak
<https://celestrak.org/NORAD/elements/gp.php?GROUP=STATIONS&FORMAT=XML>
- JSON format (pretty print) for all objects from the last Starlink launch using International Designator 2020-025
<https://celestrak.org/NORAD/elements/gp.php?INTDES=2020-025&FORMAT=JSON-PRETTY>
- JSON format for all objects with COSMOS 2251 DEB in their name
https://celestrak.org/NORAD/elements/gp.php?NAME=COSMOS-2251_DEB&FORMAT=JSON
- CSV format for GPS Ops list found on Celestrak
<https://celestrak.org/NORAD/elements/gp.php?GROUP=GPS-OPS&FORMAT=CSV>
- CSV format for GEO Protected Zone objects
<https://celestrak.org/NORAD/elements/gp.php?SPECIAL=GPZ&FORMAT=CSV>
- JSON format (pretty print) format for GEO Protected Zone Plus objects
<https://celestrak.org/NORAD/elements/gp.php?SPECIAL=GPZ-PLUS&FORMAT=JSON-PRETTY>
- JSON format (pretty print) format for Potential Decays
<https://celestrak.org/NORAD/elements/gp.php?SPECIAL=DECAYING&FORMAT=JSON-PRETTY>

There are also queries to show the first or last GP data available. For example, if you wanted to see when the first 18 SDS GP data became available for the Transporter-11 mission (2024-149), which was launched 2024-08-16, you might use:

- <https://celestrak.org/NORAD/elements/gp-first.php?INTDES=2024-149&FORMAT=JSON-PRETTY>

This type of query is also useful for getting the first data for objects in a debris event, which might take days or even months to get cataloged:

- <https://celestrak.org/NORAD/elements/gp-first.php?NAME=COSMOS-2251-DEB>

And we use the last GP query in things like our table of Lost Objects (objects which should have GP data but for which none was found in the last 30 days):

- <https://celestrak.org/satcat/lost.php>

as well as in our table of Recently Decayed Objects:

- <https://celestrak.org/satcat/decayed-with-last.php>

These are custom versions of our general table queries, which layout basic information for both the GP and SupGP data, in an interactive table. These table queries use the same structure as the GP queries. Here the FORMAT specification is used to define the format of any data linked to the table. The content (columns) may vary, depending on the focus of the data.

Examples:

- XML format for the Stations list found on Celestrak
<https://celestrak.org/NORAD/elements/table.php?GROUP=STATIONS&FORMAT=XML>
- CSV format for GEO Protected Zone objects
<https://celestrak.org/NORAD/elements/table.php?SPECIAL=gpz&FORMAT=CSV>

These table queries can also include a variety of flags to further customize the table for specific uses.

Flags:

- BSTAR: Show the BSTAR value instead of eccentricity. Of course, this is only useful for LO objects where BSTAR is computed.
- SHOW-OPS: Show the operational status flag following the name of the satellite.
- OLDEST: Show the only objects with data older than 3.5 days old, sorted from oldest to newest.
- DOCKED: Show only those objects docked to another object (e.g., ISS or CSS).
- MOVERS: In the Active Geosynchronous list (a customized table), show only those objects drifting more than 0.1° per day.

Examples:

- Celestrak uses SHOW-OPS and BSTAR to determine changes in Starlink operational status. Sorting on BSTAR (descending) and filtering on [+] shows when Starlink satellites are having their orbits lowered for disposal or are decaying. Sorting on BSTAR (ascending from negative values) and filtering on [P] can show when partially operational satellites may have been recovered and are performing orbit-raising.
<https://celestrak.org/NORAD/elements/supplemental/table.php?FILE=starlink&SHOW-OPS&BSTAR>
- Celestrak uses OLDEST with the Active satellites list to only show those satellite's whose GP data is more than 3.5 days old (normally less than 50) instead of loading data for all 10,000+ satellites. That helps focus attention on which of those satellites might have SupGP data to help relocate them.
<https://celestrak.org/NORAD/elements/table.php?GROUP=active&SHOW-OPS&OLDEST>
- Celestrak uses DOCKED with the Active satellites list to keep track of what going on with the growing set of objects docked to space stations or other satellites.
<https://celestrak.org/NORAD/elements/table.php?GROUP=active&SHOW-OPS&DOCKED>
- Celestrak uses MOVERS with the Active Geosynchronous satellites list to keep track of the small set of satellites being sent to GEO Graveyard, moving east or west to relocate, or which might have died in GEO.
<https://celestrak.org/NORAD/elements/table-geo.php?MOVERS>

And for information on how to query SupGP data, see [How to Perform SupGP Queries](#).

For Software Developers

Software developers looking for code to input or output these formats in a variety of languages are invited to check out the [Space Data Standards](#) web site developed by our partners at [Digital Arsenal](#).

There is code to support C++, Kotlin, Java, C#, Go, Python, JavaScript, TypeScript, PHP, Dart, Lua, Lobster, Swift, and JSON Schema. There are also examples converting TLEs to OMMs in XML, KVN, JSON, CSV, and FlatBuffers. And there is a GitHub repository for the code that allows users to submit suggested changes.

Summary

Providing GP data in the OMM-compatible formats provides a way forward for all software developers to continue to support using SGP4 in their applications and add support for 9-digit catalog numbers. In addition, it eliminates the Y2K problem still coming NLT 2057 by using the [ISO 8601-1 \(WD\)](#) date and time standard and also supports the use of Unicode characters for satellite names from non-English languages. Eventually, it will also avoid limitations with 3-digit International Designators, as well (we had [315 successful launches in 2025 alone](#)).

FAQs

Q: Why don't we just modify the current 5-digit catalog numbers to include letters to increase the range of objects that can be represented?

A: There have been numbering schemes suggested that would extend the range of catalog IDs that would fit in a 5-character field of a TLE. The easiest would be to extend the current 'numbering' by allowing the leading character to go from 0-9 and then A-Z. At best, this would allow for tracking 360,000 objects (assuming you don't discard I and O, as has been suggested by some). That could work for the Space Fence, but ignores other potential large increases in objects tracked due to the development of large constellations (which currently propose as many as 100,000 new satellites) or additional large debris events that might occur due to the collision of large uncontrolled rocket bodies.

Even allowing all 5 characters to go from 0-9 and then A-Z would only allow 60,466,176 catalog IDs, but 18 SPCS is already assigning catalog numbers in their new analyst sat range of 7995xxxxx (or over 799,500,000). So there would be no way to map these 9-digit catalog numbers to 5-character IDs.

And the ability to fit new characters into a field does not make the problem of interpreting the change in software go away, any more than the change in interpretation of the 2-digit year did for Y2K. All the code using these catalog numbers will have to be updated to change their interpretation, with a cascading set of implications. Including letters means the field will no longer be able to be simply validated by verifying that it is an integer and integer comparisons will no longer be possible. For example, some TLEs use leading zeros in the 5-digit field while others do not. But a value of 00964 parses as an integer the same way as 964.

Every software developer will need to update their code to adapt, so this is an opportunity to do that in a way that provides future flexibility and no longer relies on the limitations of fixed formats.

Q: Why do we have to use formats that are so verbose?

A: While it is possible to use formats that are less verbose, they do not provide the framework to ensure future interoperability that an international standard like the OMM provides. Software developers—particularly those developing to support systems for use in satellite operations, ensuring safety of flight, or national security—are *strongly encouraged* to use the recommended OMM XML standard. Others that do not support critical functions like these may choose the CSV or JSON formats based on the OMM standard (but not currently part of that standard), although there is always the possibility that these could change. Celestrak has worked hard to ensure that doesn't happen, going back almost 35 years now, and will endeavor to make changes in a way that maintain backward compatibility whenever possible. Note that **Celestrak uses the CSV format behind the scenes**, since it is typically smaller than TLE-formatted data, easy to visually inspect or edit, easy to parse, and readily loads into any spreadsheet software—so we aren't going to change anything in that format unless absolutely necessary. **The JSON format is 3 times the size of the comparable CSV data**, due to its redundant structure (*XML and KVN are much worse*).

But the reality is that using a full XML version of the OMM for a single object takes about 1,200 characters (including all the overhead XML formatting for a set of OMMs) compared to the 168 characters of a three-line element set. That is a factor of 7 larger. When I first started Celestrak in 1985, I had a 1200-baud (120 Bps) modem on the system for a single user at a time. Today, Celestrak has a 1-Gbps connection that can support as many as 800,000 unique users a day (demonstrated) and my home Internet service allows up to 500 Mbps—that's a factor of over 500,000 times faster. And my hard disk at the time was a whopping 5 MB—I have eight external 24-TB & 18-TB drives on my home system (each that cost a tiny fraction of that 5-MB one) and Celestrak has 400 GB of SSD storage. That's a factor of 80 to 1,000 times as much storage. And we will continue to see similar advancements in bandwidth and storage that make these differences irrelevant.

And the long-established (February 1998) XML format has extensive software support to allow easily ingesting the OMM XML data.

FAQ Addendum

Added 2024 Aug 30
Updated 2026 Mar 26

Q: Why am I getting blocked trying to download GP data?

A: Celestrak only checks for new GP data once every 2 hours, so there is no need for you to check more often. In fact, when you do, that uses limited resources needed to support hundreds of thousands of unique users on Celestrak each day. Because some users, if left unchecked, will download the same file every minute of the day (that's 1,440 times or 120x the update rate), **every day**, we have had to implement limits, which are enforced with temporary blocks.

If the IP address for you (or your process) is being blocked, Celestrak sends a custom HTTP 403 error message explaining why you are being blocked:

403 - Forbidden: Access is denied.

Access from IP address xxx.xxx.xxx.xxx was blocked for celestrak.org/NORAD/elements/gp.php?CATNR=25544&FORMAT=TLE.

We have detected excessive downloads for files in the /NORAD/elements directory and access has been temporarily blocked. Access will be automatically restored once the excessive downloads have ceased for 2 hours.

Please note that orbital data files are only checked for updates every 2 hours and most orbital data only updates 2-3 times a day (or less). Please check your scripts to ensure they are operating properly.

Please contact ts.kelso@celestrak.org if you have questions.

Of course, that means you (and your processes) need to be checking for error messages. If you check the query being blocked in your browser (from the same IP address that your process is using), you should immediately see what's going on. If you still don't understand why you're being blocked, you can send me a screenshot, like the one above, which includes the IP address, and I will look into it. I often help users who think they are only making a small number of requests realize their process hit an unexpected response and just started hammering the system.

Your process can avoid this situation by checking for a successful response (an HTTP 200) and being prepared to handle unexpected responses. If some other response is received, your process should stop and report the problem to a human. **In particular, if you receive an HTTP 403 or 404 error, the response is not going to change by repeating the request and can result in your IP address being put in the firewall.** On Celestrak, we follow this approach when downloading data from any other sites. When a serious problem is encountered, our processes actually send an SMS (text) message to ensure quick attention. And each of our processes maintains an easily accessible log file on our Dashboard to clearly report what happened.

Failing to have automated processes check for errors can not only waste Celestrak's limited resources, it can cause users to blindly continue to use data that hasn't been updated for days, months, or even years (yes, we have seen all of this).

For example, Celestrak changed the primary domain to <https://celestrak.org> years ago when we became a non-profit on 2021 Apr 26. That means if you use the /NORAD/elements directory, Celestrak tries to redirect your query to the .org domain and sends an HTTP 301 (Moved Permanently) response. Your browser knows how to handle that, but your process may not, causing it to hammer away until we block it. Once the system sends more than 1,000 HTTP 403 errors (and now 301 and 404 errors, too) to an IP address in a day (yes, that happens almost every day), that IP address is put into the firewall and requires manual review to find and remove it.

To put this in perspective, let's look at a snapshot from 2024 Aug 29 (yesterday). Of the 864,374 successful accesses on the site (HTTP 200), 571,795 of them started out on the .com domain and received an HTTP 301—almost 3.5 (now 6.3) years after the change. Not only does that mean Celestrak has to execute (and log) twice as many queries, it could mean those users aren't getting any data.

On 2023 Dec 28 (8 months ago), we removed a number of legacy static .txt files, which only use the TLE format, in an effort to get users to prepare for running out of 5-digit catalog catalog numbers in the main part of the SATCAT ([see notice on Bluesky](#)). **If you thought that happens at 99999, you may be surprised to discover that is not true (see notice on Bluesky with updates).** When we run out of 5-digit catalog numbers at 99999, new data will not be able to be created using the TLE format. In the meantime (just yesterday), we saw 30 of these deleted legacy files requested between 194 and 3,301 times each (42,082 times total). These users have likely received no data for as much as 8 months.

We finally removed ALL of these legacy files on 2024 Dec 24, following yet more cases of excessive or malicious behavior ([see notice on Bluesky](#)). And after the latest 18 SDS/Space-Track data outage 2025-08-21-24, where we got hammered by users repeatedly accessing Celestrak trying to get fresh GP data (which we get from Space Track)—many of them using deprecated queries—**we now set a limit on HTTP errors (301, 403, or 404) of 50 in a 2-hour period, at which point the IP address is sent to the firewall.** These changes aren't intended to be punitive, rather they have been made to get users' attention to the impending end of the TLE format and to encourage users to adapt their processes to respect Celestrak's resource limitations and the other users who do.

All of this started out to encourage users to prepare for the near future, which is why this documentation is here. In fact, Celestrak already provides GP data for USSF Space Fence analyst objects that use 6-digit catalog numbers. You can see that toward the end of [this table](#) but if you click on the [link for the TLE-formatted data](#) in the header, you will notice it does not include any of the 27xxxx catalog numbers (it only shows 8xxxx catalog numbers). Using a format like CSV or JSON will include **all of the data**. And if you look closely at the SupGP data for a recent Starlink launch, you will notice we are already using the 18 SDS 9-digit launch nominators catalog numbers (in the 799xxxxxx range). The same will be true for future Transporter and Bandwagon launches.

The bottom line here is that these blocks are in place to get users' attention that their processes are likely not working as expected. It is unlikely that anyone is manually requesting hundreds or thousands of downloads a day. But since we don't have user accounts, we can't just send you a message. So, we use these progressive steps to (hopefully) get your attention, so that you aren't left blissfully unaware that you may not be getting new data or of impending changes to data formats.

Q: How can I avoid getting blocked?

A: Now that you know why you are getting blocked, it's actually pretty easy to solve the problem.

First, turn off the process causing the issue. Once you do this, the temporary blocks will be automatically removed within 2 hours.

Next, modify your process to use the latest data you downloaded by default. If nothing else, this step will allow your process to continue working in the event of temporary Internet issues.

Finally, add a step before using the latest data to check the data file's timestamp to see if it is more than 2 hours old. If it is, re-download the data to that file and proceed to use it. Otherwise, just use the latest data. Pretty simple. Note that when set up this way, if you are a software developer testing your code, the process is still only going to download the data only once every 2 hours (at most). Oh, and only download the data you need, when you need it. There really isn't any need to download after and only Celestrak update, since the 18 SDS GP data only updates 2-3 times a day. You can see that in the second graph [here](#). Zoom in to 1m (1 month). Updates occur when the mean age decreases. This page is also very helpful for seeing why some (or all) of the data doesn't seem to be updating.

And be sure your process is checking for error responses (e.g., HTTP 301, 403, 404, or 500) and stopping additional queries when these are detected and reporting them to a human for investigation.

Q: How else can I help Celestrak make the most out of its limited resources?

A: First and foremost, only download the data you need, when you need it (are ready to use it). Back in the day (circa 20 years ago), many processes were written to harvest all of those legacy static .txt files, many times a day. Often that data just took up disk space and never got used. Plus, downloading data every 12 hours, just so you have it, likely only meant it would be six hours old (on average) when you went to use it. Now that Internet speeds are faster and connections are more reliable, it's better to just grab the data when you need it, which can also randomize when that occurs (which spreads out the load on Celestrak). And don't be that person that still needs to download all of the data—including that for the old Iridium satellites that are all long dead, uncontrolled, and no longer generate flares—oh yeah, and because that file is now gone.

UPDATE: Since 2026 Feb 16, we have seen bandwidth usage jump from ~125 GB/day to ~330 GB/day (Mar 17) for roughly the same number of unique IP addresses. That means we will blow through our 6-TB monthly bandwidth in just over two weeks. As a result, we are now forced to implement bandwidth usage checks. Analysis of our logs show only a very small percentage of users (~0.3%) are using more than 100 MB/day and they are all doing things like downloading large files many times more often than they are updated. We are not going to pay extra money to allow users to wantonly disregard our requests to respect our resource limitations. If you are using more than 100 MB/day you can expect that your IP address may end up in the firewall.

UPDATE: It appears that setting a 250-MB daily limit to discourage the 100–150 users a day who feel no need to respect our resource limits hasn't achieved our goal, so Celestrak will now (as of 2026 Mar 26) simply enforce the one-download-per-update policy for all users, starting with the Active and Starlink GROUPs. These requests are the overwhelming request of those wasting our bandwidth and slowing performance for everyone. The first request will work fine, but the second will return something like this (with an HTTP 403 response) until the GP data updates again:

GP data has not updated since your last successful download of GROUP=active at 2026-03-26 08:10:22 UTC.

Data is updated once every 2 hours.

Continued excessive requests—whether you get data or not—can still result in your IP address being sent to the firewall.

You can avoid this result by carefully considering how much data you need and not downloading new data more than once every 2 hours. There is no need to download the list of active satellites and the list of all Starlink satellites, since the latter is a subset of the former. There is no reason to download all of the GROUPs, since these are intended to help users only download the smaller sets of satellites they need (e.g., amateur radio or visible). There is no need to download data until you are ready to use it, which ensures you have the latest data you need when you do.

Along these same lines, the most common abuse we see is for users to download the list of active satellites every ten minutes or less. You know that we will start enforcing only one download per update soon—starting with larger files—if other efforts to reduce excessive downloads are not successful. When that happens, users will see a message stating the data has not updated since their last successful download instead of receiving data.

Of course, if you aren't a software developer or are using an application written by someone else, and are seeing problems with getting blocked, please be sure to pass this information along to them so that they can take the necessary steps to avoid it. That not only helps you, but others using the same software.

Final Note

Please realize that Celestrak makes these resources freely available to all users, but that doesn't mean it doesn't cost us anything to do so. Even though we get millions of unique users on the site each month, very few users—including those who profit from our efforts—donate anything to help us cover our operations (I pay for all of that out of my own pocket). If you value what we do and want to ensure that these services continue to be available in the future, please consider starting a [monthly donation today](#). We (the Celestrak community) need to be able to cover not only operations and development, but hiring staff to cover everything we do (including things like system administration, to ensure reliable performance, and cybersecurity).